Radu Dehel

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Skills

Programming Languages: Python, MATLAB, C/C++, Java, SQL, VHDL
Technical Skills: KiCad, Solidworks, LTspice, Fusion 360, EasyEDA, UART/I2C/SPI, OOP, STM32, ESP32, Arduino, Multithreading, MQTT, Qt Designer, DMM, Oscilloscopes, Semiconductor Manufacturing, EUV Lithography
Libraries/Tools: OpenCV, Pandas, Numpy, Scikit-Learn, MATLAB Image Processing Toolbox, MATLAB Signal Processing Toolbox, Paho-MQTT, Selenium, Qt, MySQL, Git/Github, JIRA

Experience

Electronics Engineering Intern January 2026 – May 2026
ASML *Veldhoven, the Netherlands*

- Designed a 4 layer PCB in KiCad to control and emulate 42 temperature sensors with a 1kHz update rate at 20-bit resolution, enabling safe closed-loop thermal control testing and reducing $500k+ in hardware testing costs.
- Analyzed SPI signal integrity using an oscilloscope to identify ringing, overshooting and rise/fall times, implemented a dual Schmitt trigger stage to eliminate distortions on a custom PCBA.
- Implemented TVS-diode ESD protection circuitry on all digital and analog I/O lines and integrated hot-swappable capabilities to prevent damage to sensitive components.
- Developed an automated hardware-in-the-loop (HIL) test bench to characterize sensor simulator performance, measuring thermal stability, noise floor, and dynamic response using MATLAB, C (STM32) and a precision DMM.

Mechatronics Engineering Intern June 2025 – August 2025
TITAN Haptics *Mississauga, Ontario*

- Designed a PCB for a motor testing platform, integrating microcontrollers, H-bridge motor control, Li-ion charging, and power protection circuits.
- Conducted motor reliability testing and data analysis in Python to identify causes of performance loss (thermal cycling, coil resistance, magnet degradation, frequency spectrum).
- Designed a motor test fixture in Fusion 360, integrating accelerometer, Gauss meter and thermal sensors to better characterize performance of motors.

EUV Lithography Data Science Intern (Overlay/Alignment) July 2024 – March 2025
ASML *Veldhoven, the Netherlands*

- Contributed to ASML's overlay MATLAB toolbox (used by over 4,000 engineers) by developing data analysis tools and scripts for processing, visualizing, and analyzing wafer overlay data for EUV machines.
- Designed vector field filtering and fitting algorithms in MATLAB to identify part wear trends in wafers, clamps, and stage components to predict optimal replacement times.
- Applied time series analysis and cross-data correlation on wafer metrology and overlay data to identify the root cause behind a 35% decline in overlay KPI.
- Conducted research across EUV lithography systems, identifying component-level impacts on alignment KPIs and providing insights for future hardware improvements.

Aerospace Image Sensor Technologist Intern September 2023 – December 2023
Teledyne DALSA *Waterloo, Ontario*

- Designed optical testing fixtures involving the configuration of light sources, optical power meters, and FPGA boards, while integrating LabVIEW and Matlab code to guarantee precise results.
- Created MATLAB test code for measuring dynamic range, saturation and quantum efficiency for CMOS sensors.
- Designed Python script to automate the testing of analog to digital converters on image sensor chips.
- Developed a Python image processing script that finds defects in optical filters using OpenCV and NumPy.

Hardware Design Team Leader - Lithography & Stage Team December 2024 – Present
Waterloo Hacker Fab *Waterloo, Ontario*

- Designed a piezo stage and controller PCB for precision positioning of wafer during lithography using KiCad, LTspice and SolidWorks; implemented op-amps, analog filters, and DACs to achieve 1 um displacement resolution.

Education

University of Waterloo Waterloo, Ontario
Candidate for Bachelor of Applied Science, Honours Electrical Engineering *September 2022 – Present*